

5. System Response and LTI Properties

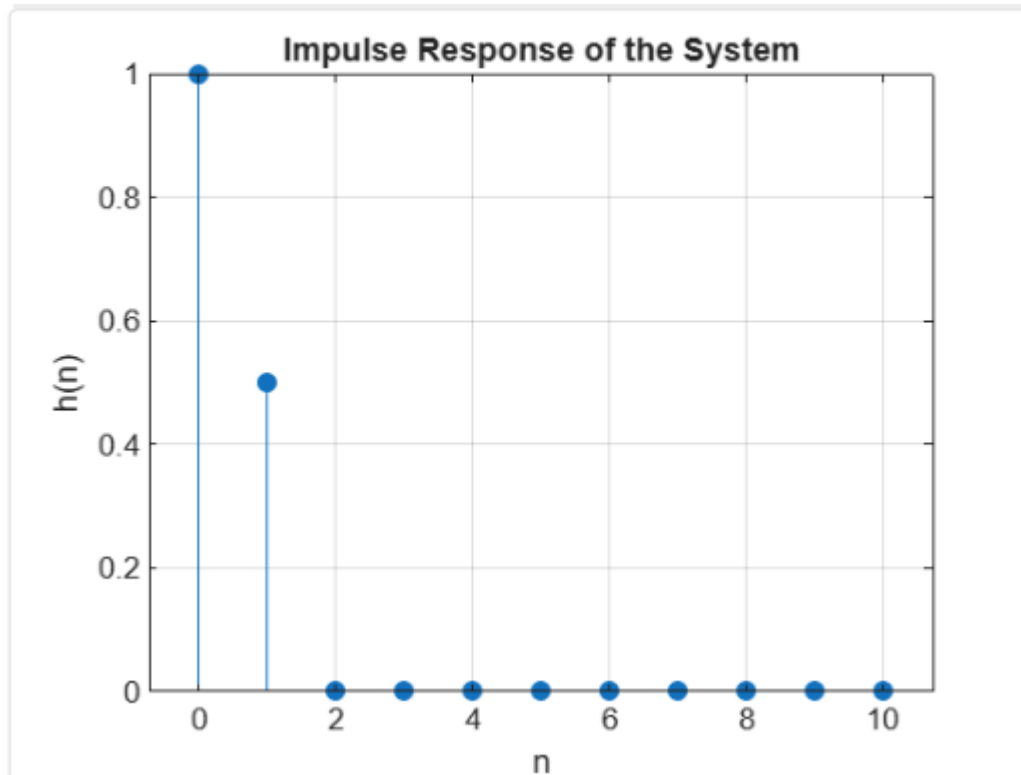
Experiment 5.1

% Impulse Response of the System

```
clc;
clear;
close all;
% Sample index
n = 0:10;
% Unit impulse input
x = [1 zeros(1,10)];
% System:  $y(n)=x(n)+0.5x(n-1)$ 
b = [1 0.5];
a = [1];
% Impulse response
h = filter(b,a,x);
% Display result
disp('Impulse Response h(n):');
disp(h);
% Plot
figure;
stem(n,h,'filled');
grid on;
title('Impulse Response of the System');
xlabel('n');
ylabel('h(n)');
```

Impulse Response $h(n)$:

1.0000 0.5000 0 0 0 0



Experiment 5.2

% Step Response of the System

```
clc;
```

```
clear;
```

```
close all;
```

% Sample index

```
n = 0:10;
```

% Unit step input

```
x = ones(1,length(n));
```

% System: $y(n)=x(n)+0.5x(n-1)$

```
b = [1 0.5];
```

```
a = [1];
```

```
% Step response
y = filter(b,a,x);
% Display result
disp('Step Response y(n):');
disp(y);
% Plot
figure;
subplot(2,1,1);
stem(n,x,'filled');
grid on;
title('Unit Step Input u(n)');
xlabel('n');
ylabel('Amplitude');
subplot(2,1,2);
stem(n,y,'filled');
grid on;
title('Step Response of the System');
xlabel('n');
ylabel('y(n)');
```

Experiment 5.3% Verification of Linearity Property of LTI System

```
clc;
clear;
close all;
```

% Sample index

n = 0:10;

% Input signals

x1 = [1 2 3 4 5 6 7 8 9 10 11];

x2 = [11 10 9 8 7 6 5 4 3 2 1];

% Constants

a = 2;

b = 3;

% Combined input

x = a*x1 + b*x2;

% System: $y(n)=x(n)+x(n-1)$

sys = @(x) filter([1 1],1,x);

% LHS = $T[a*x1+b*x2]$

LHS = sys(x);

% RHS = $a*T[x1]+b*T[x2]$

RHS = a*sys(x1) + b*sys(x2);

% Display results

disp('LHS = ');

disp(LHS);

disp('RHS = ');

disp(RHS);

% Verification

if isequal(LHS,RHS)

```

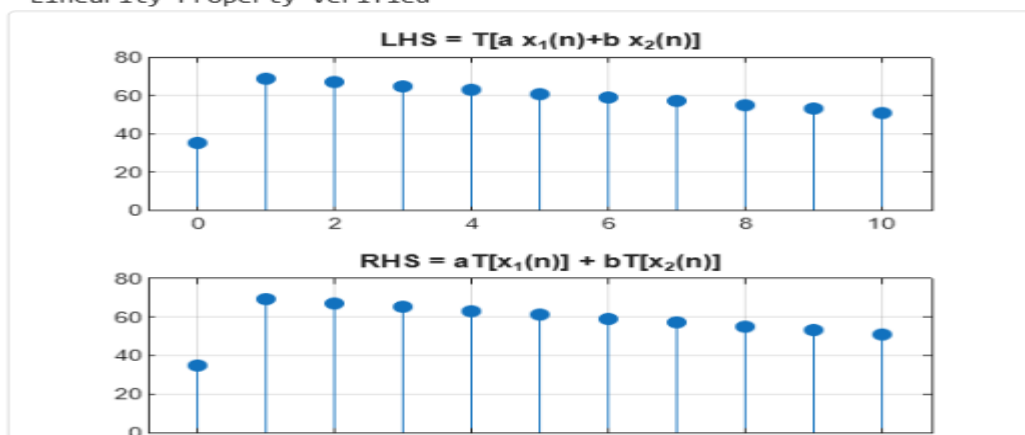
disp('Linearity Property Verified');
else
disp('Linearity Property Not Verified');
end
% Plot
figure;
subplot(2,1,1);
stem(n,LHS,'filled');
grid on;
title('LHS = T[a x_1(n)+b x_2(n)]');
subplot(2,1,2);
stem(n,RHS,'filled');
grid on;
title('RHS = aT[x_1(n)] + bT[x_2(n)]');

```

```

LHS =
    35    69    67    65    63    61    59    57    55    53    51
RHS =
    35    69    67    65    63    61    59    57    55    53    51
Linearity Property Verified

```



Experiment 5.4

% Analysis of Discrete-Time System

```
clc;
clear;
close all;
% Sample index
n = 0:20;
% Input sequence (Unit Step)
x = ones(1,length(n));
% System coefficients
b = [1];
a = [1 -0.5];
% System response
y = filter(b,a,x);
% Display output
disp('Output Sequence y(n):');
disp(y);
% Plotting
figure;
subplot(2,1,1);
stem(n,x,'filled');
grid on;
title('Input Signal x(n)');
```

```

xlabel('n');
ylabel('Amplitude');
subplot(2,1,2);
stem(n,y,'filled');
grid on;
title('Output Signal y(n)');
xlabel('n');
ylabel('Amplitude');

```

